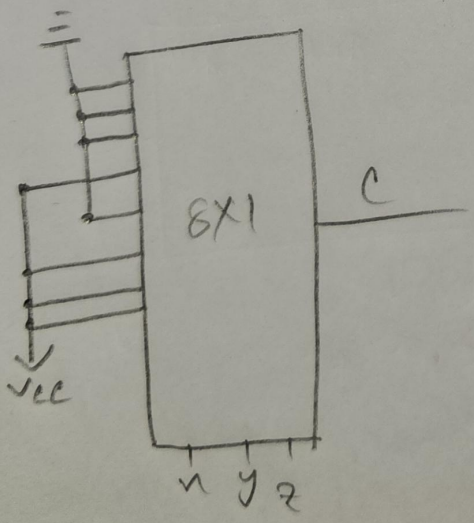
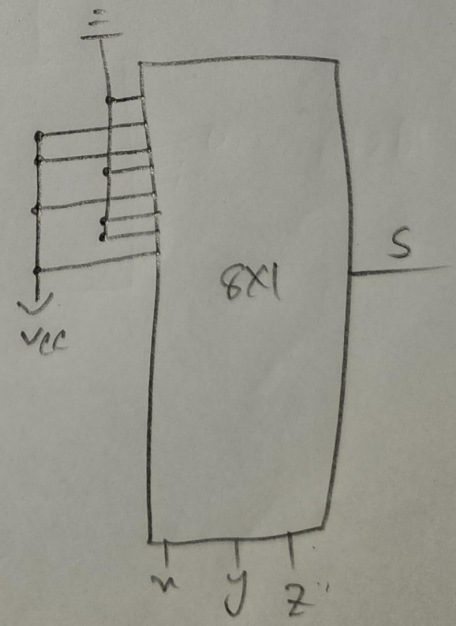
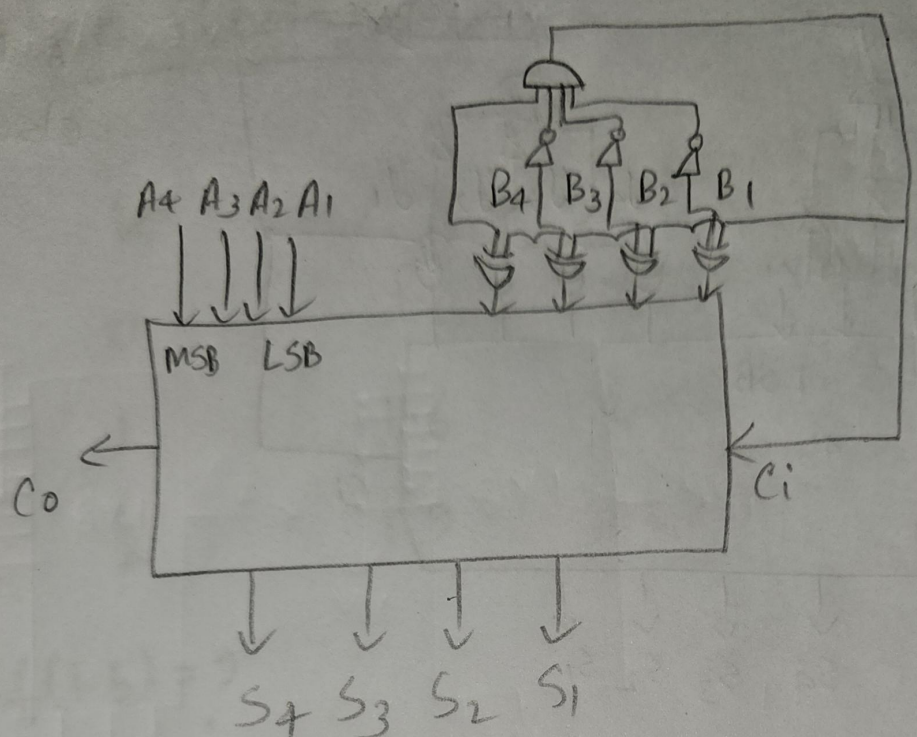
*) full adder using two 8X1 mux

x	y	z	C	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

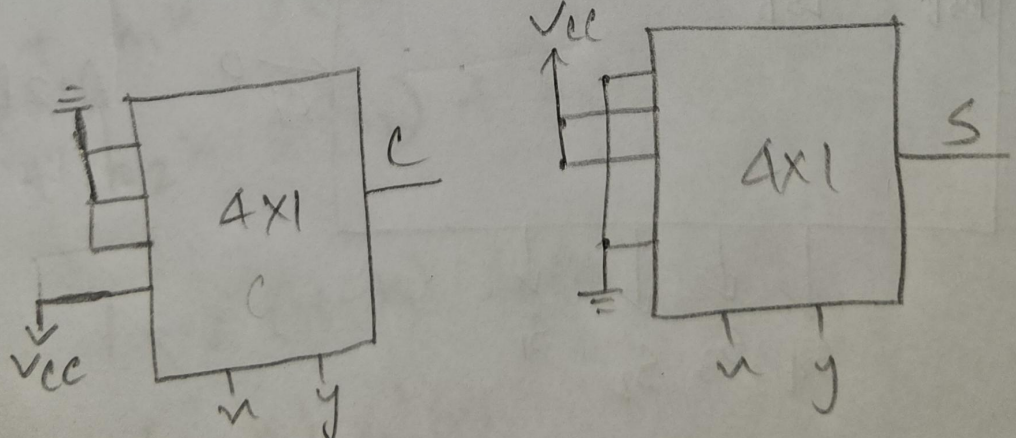






*) half adder with two 4:1 Mux

$C = xy$, $S = x'y + xy'$



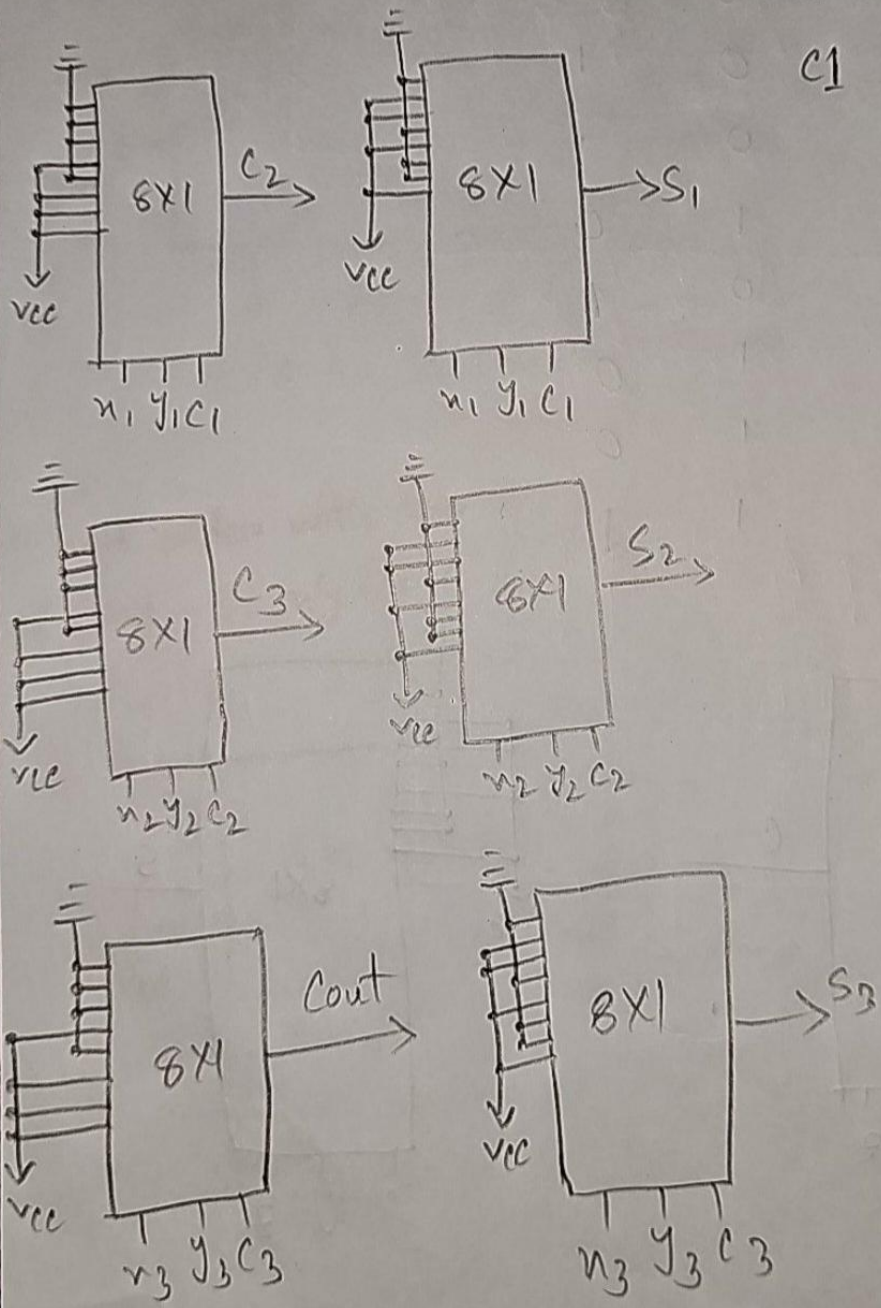
*) 3bPA using 8X1 Mux

x_3 x_2 x_1

y_3 y_2 y_1

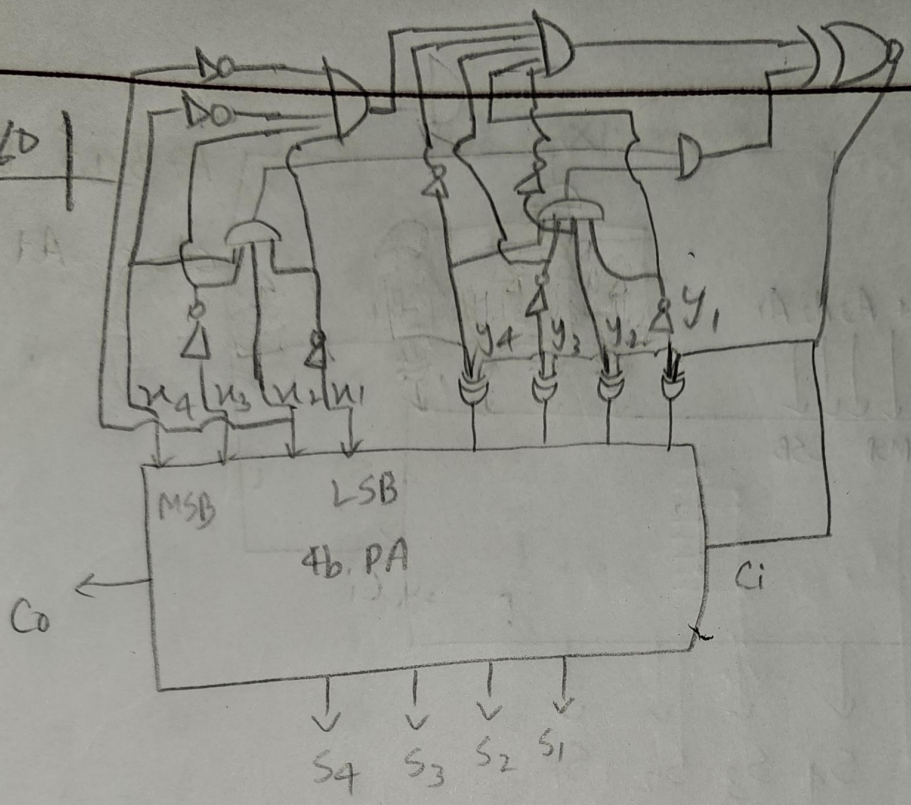
MSB LSB

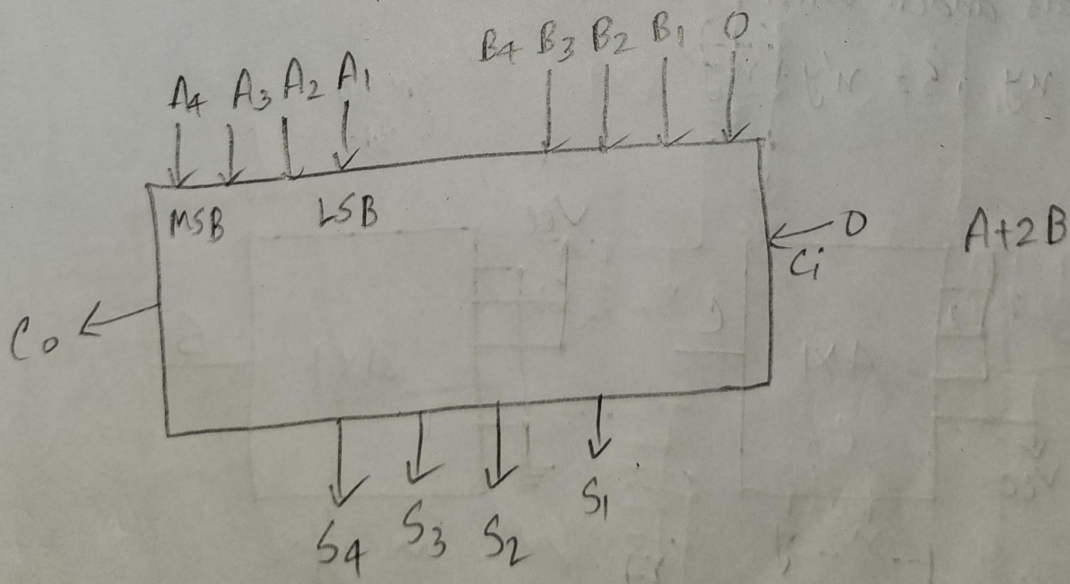
C1



CSE 260

A+B if A,B/10=0
else A-B



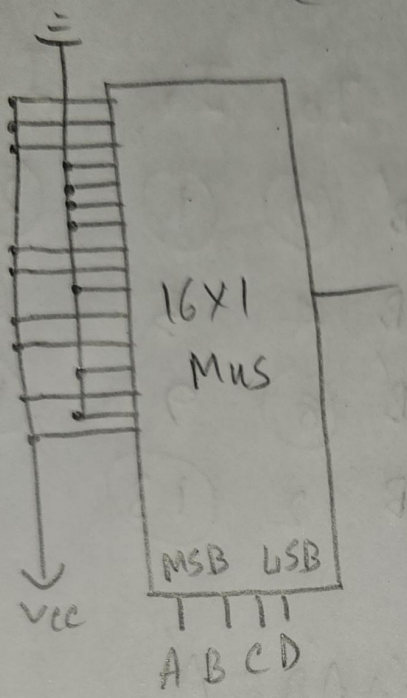


A+2B

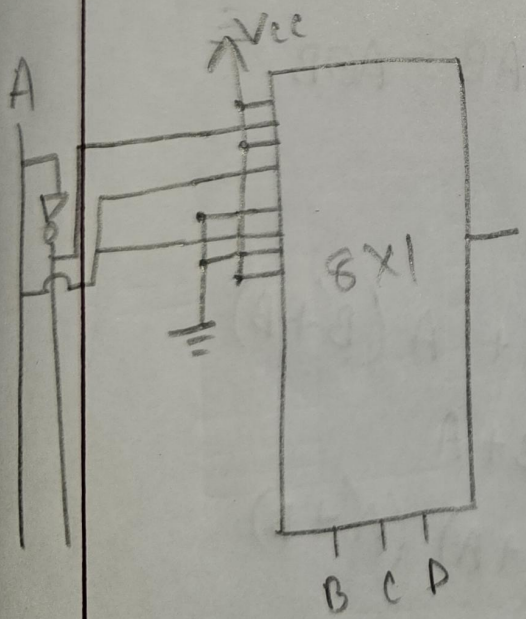
*) city 'GG'

	F_3	F_2	F_1	F_0	A_2	A_1	V
	0	0	0	0	X	X	0
	0	0	0	1	0	0	1
	0	0	1	0	0	1	1
	0	0	1	1	0	0	1
4	0	1	0	0	1	0	1
	0	1	0	1	0	0	1
	0	1	1	0	0	1	1
3	0	1	1	1	0	1	1
8	1	0	0	0	1	1	1
	1	0	0	1	0	0	1
	1	0	1	0	0	1	1
3	1	0	1	1	0	1	1
12	1	1	0	0	1	1	1
3	1	1	0	1	1	0	1
3	1	1	1	0	1	0	1
	1	1	1	1	0	0	1

*) $F(A,B,C,D) = \Sigma(0,1,2,7,8,10,11,13,15)$

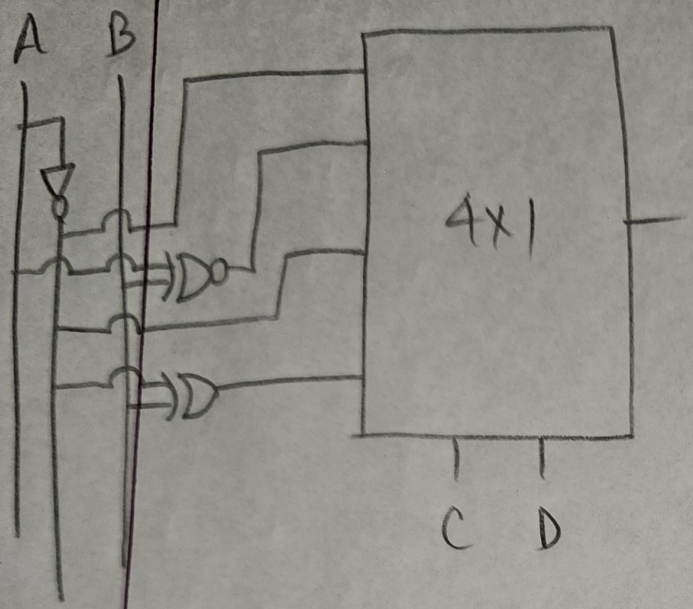


A B C D
MSB LSB



	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
A'	(0)	(1)	(2)	3	4	5	6	(7)
A	(8)	9	(10)	(11)	12	(13)	14	(15)

A'+A
= 1



	I_0	I_1	I_2	I_3
$A'B'$	(0)	(1)	(2)	3
$A'B$	4	5	6	(7)
AB'	(8)	9	(10)	(11)
AB	12	(13)	14	(15)

$I_0 = A'B' + A'B = A'$

$I_1 = A'B' + AB = A \odot B$

$I_2 = A'$

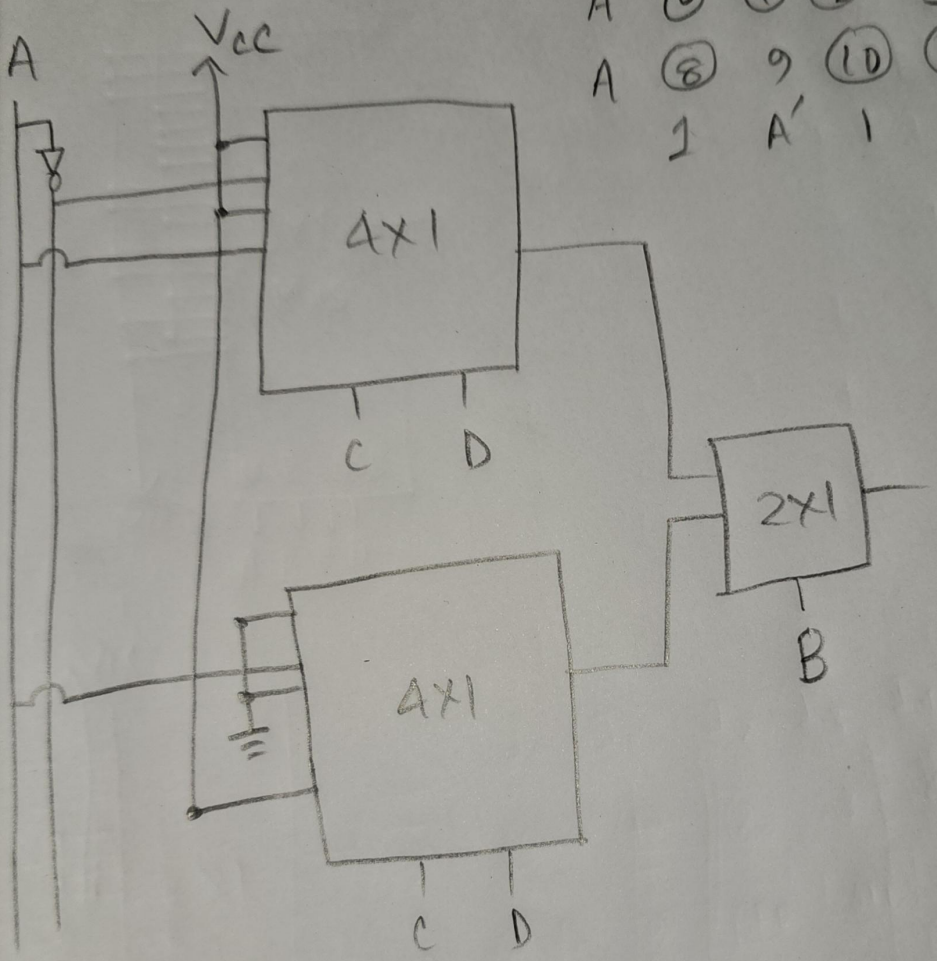
$I_3 = A'B + A(B' + B)$

$= A'B + A$

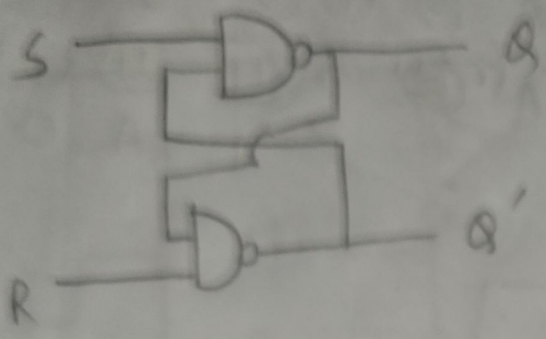
$= (A' + A)(A' + B)$

$= A' + B$

	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
A'	(0)	(1)	(2)	3	4	5	6	(7)
A	(8)	9	(10)	(11)	12	(13)	14	(15)
	1	A'	1	A	0	A	0	1



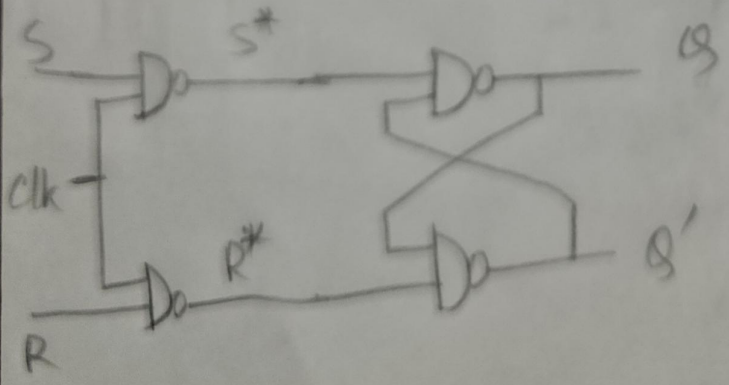
SR Latch



SR Latch

S	R	Q
0	0	Not used
0	1	1
1	0	0
1	1	Memory

SR FF



S	R	S*	R*	Q
0	0	1	1	Memory
0	1	1	0	0
1	0	0	1	1
1	1	0	0	Not used

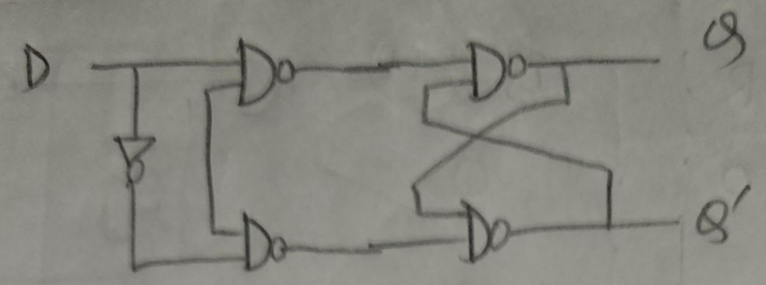
SR FF characteristic table

Q(t)	S	R	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	Not used
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	Not used

excitation table

Q	Q^+	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

D FF



T T

D	Q
0	0
1	1

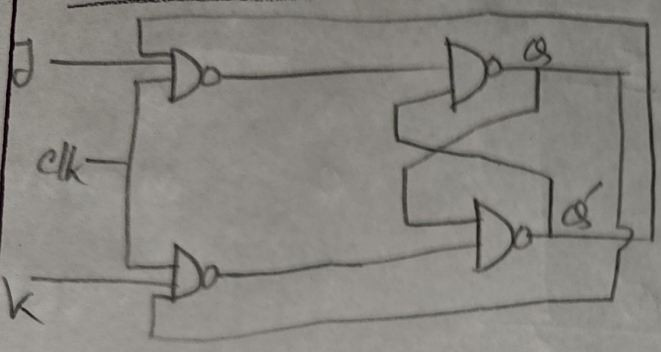
Characteristic T

Q(t)	D	Q(t+1)
0	0	0
0	1	1
1	0	0
1	1	1

Excitation table

Q	Q^+	D
0	0	0
0	1	1
1	0	0
1	1	1

Jk FF

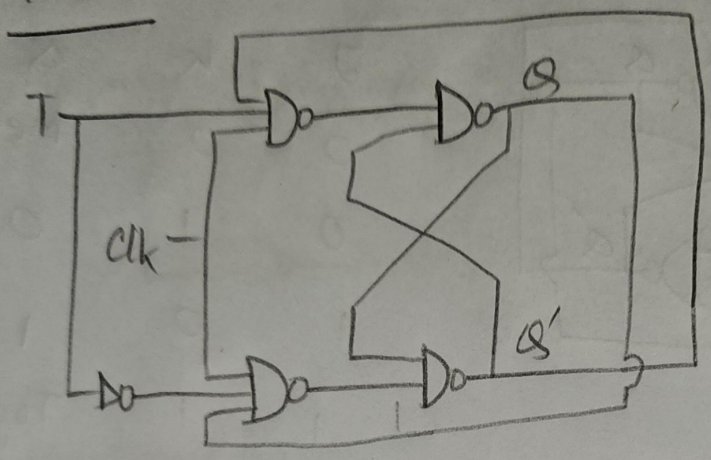


J	k	Q
0	0	Memory
0	1	0
1	0	1
1	1	Toggle

Q(t)	J	k	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

Q	Q^{t+1}	J	k
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

T FF



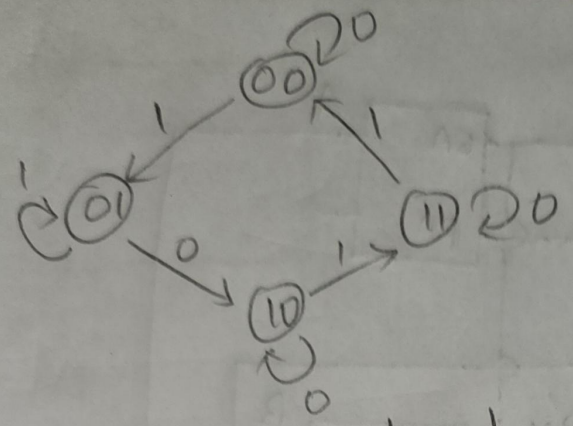
TT

T	Q
0	Memory
1	Toggle

Q(t)	T	Q (t+1)
0	0	0
0	1	1
1	0	1
1	1	0

Q	Q^+	T
0	0	0
0	1	1
1	0	1
1	1	0

*)



SR FF Excitation table

Q	Q^+	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

A	B	x	A^+	B^+
0	0	0	0	0
0	0	1	0	1
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

SA	RA	SB	RB
0	X	0	X
0	X	1	0
1	0	0	1
0	X	X	0
X	0	0	X
X	0	1	0
X	0	X	0
X	1	0	1

A \ Bx	00	01	11	10
0	0	0	0	1
1	X	X	0	X

$SA = Bx'$

A \ Bx	00	01	11	10
0	X	X	X	0
1	0	0	1	0

$RA = Bx$

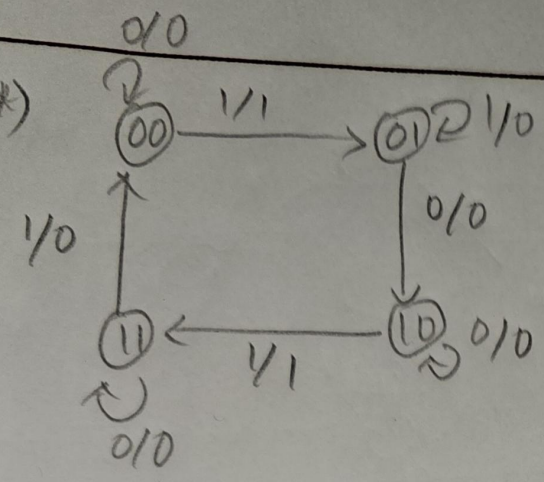
A \ Bx	00	01	11	10
0	0	1	X	0
1	0	1	0	X

$SB = B'x$

A \ Bx	00	01	11	10
0	X	0	0	1
1	X	0	1	0

$RB = Bx + A'x'$

*)

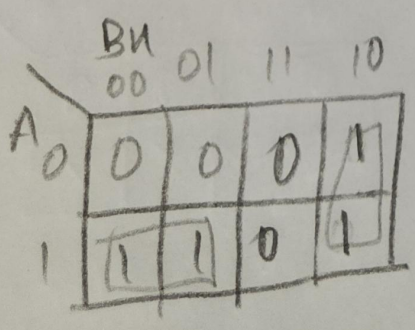


A	B	x	A^+	B^+	y
0	0	0	0	0	0
0	0	1	0	1	1
0	1	0	1	0	0
0	1	1	0	1	0
1	0	0	1	0	0
1	0	1	1	1	1
1	1	0	1	1	0
1	1	1	0	0	0

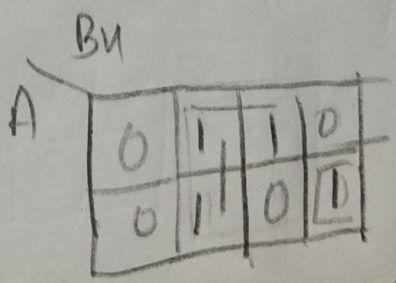
DFF

Q	Q^+	D
0	0	0
0	1	1
1	0	0
1	1	1

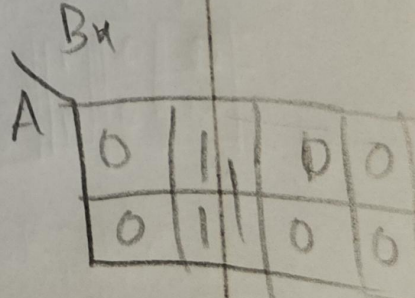
DA	DB
0	0
0	1
1	0
0	1
1	0
1	1
1	1
0	0



$DA = Bx' + AB'$

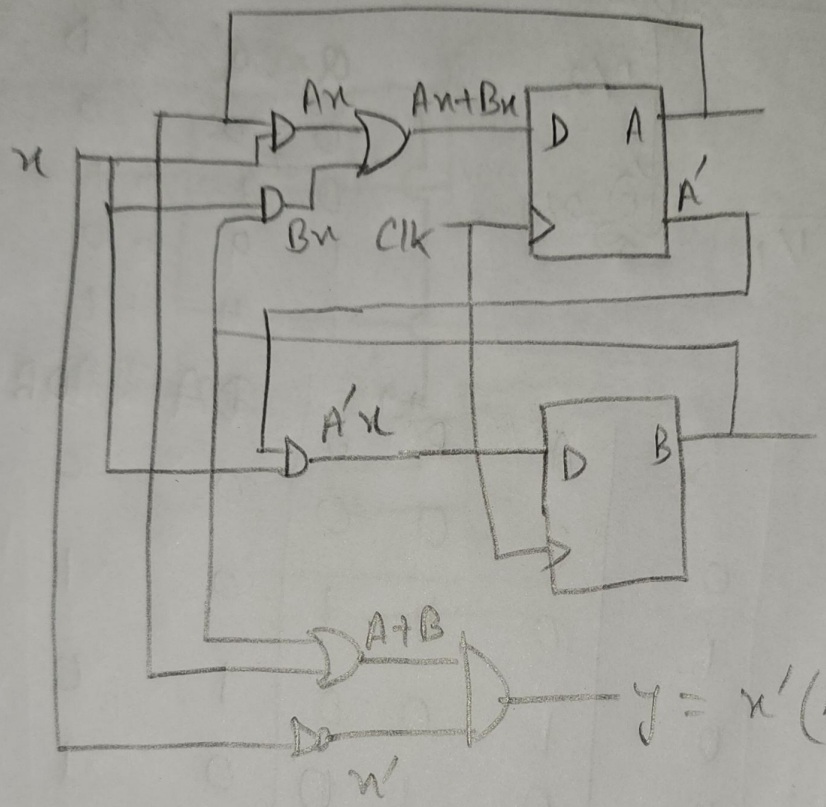


$DB = A'x + B'x + ABx'$



$Y = B'x$

D	T	T
D		Q
0		0
1		1

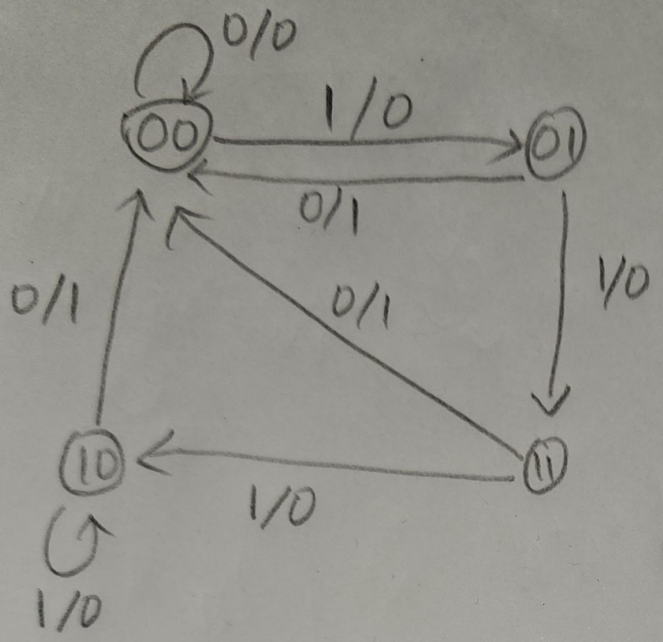


$DA = Ax + Bx$

$DB = A'x$

$y = x'(A+B)$

A	B	x	DA	DB	y	A^+	B^+
0	0	0	0	0	0	0	0
0	0	1	0	1	0	0	1
0	1	0	0	0	1	0	0
0	1	1	1	1	0	1	1
1	0	0	0	0	1	0	0
1	0	1	1	0	0	1	0
1	1	0	0	0	1	0	0
1	1	1	1	0	0	1	0


0/0
1/0
00
01
0/1
0/1
0/1
1/0
10
11
1/0
1/0